NIIT ENTERPRISE

STARROUTE | RPS

AI-NATIVE FORWARD DEPLOYED ENGINEERING (FDE)

CAPABILITY TRANSFORMATION PROGRAM

Transforming Experienced Engineers into AI-Native, Customer-Facing Forward Deployed Engineers

- Customer Centric
- AI-Native Engineering
- Enterprise Ready
- Production Focused
- Business Outcomes
- Continuous Optimization



Enabling the client's engineers to discover opportunities, build AI-powered solutions, and drive measurable business outcomes.

WHAT IS A FORWARD DEPLOYED ENGINEER?

FDEs operate at the intersection of customer context, business value, and technology execution. They own the journey from discovering opportunities to delivering production-ready solutions and driving outcomes.

FDE IN CONTEXT: DIFFERENTIATION FROM ADJACENT ROLES

Role	Primary Focus	How FDE Is Different
Developer	Implement features against defined scope	FDE **defines** the scope through discovery, hypothesis-led building and validates in customer context.
Solution Architect	Target-state design and standards	FDE moves faster into **customer-context** validation, integration, and working solution slices.
Product Manager	Prioritization and product outcomes	FDE **builds and validates** technical solution paths inside customer reality.
Consultant	Advisory and recommendations	FDE **proves outcomes** through prototypes and production-aware delivery.
Business Analyst	Requirements and process understanding	FDE **converts ambiguity** into technical experiments, integrations, and measurable outcomes.

THE FDE ROLE



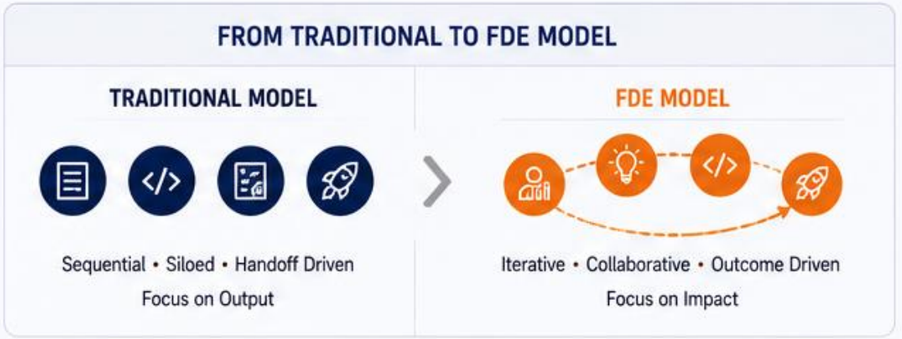
THE FDE ADVANTAGE

- Faster time to value
- Higher solution quality
- Stronger customer relationships
- Measurable business impact
- Continuous optimization

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

WHY FORWARD DEPLOYED ENGINEER MATTERS?

FDEs work in close partnership with customers across the complete solution lifecycle – from problem discovery to continuous value realization



THE FDE END-TO-END LIFECYCLE

1. DISCOVER
Understand customer context, challenges, and desired outcomes.

2. DEFINE OPPORTUNITY
Frame the problem, identify opportunities, and define success metrics.

3. DESIGN & HYPOTHESIZE
Craft solution approach, architecture options and hypotheses to validate.

4. BUILD & PROTOTYPE
Rapidly build solution slices using AI-native tools and engineering best practices.

5. INTEGRATE & DEPLOY
Integrate with enterprise systems and deploy to secure, scalable environments.

6. ADOPT & MEASURE
Enable users, measure impact, and drive adoption.

7. OPTIMIZE & SCALE
Continuously optimize solutions and scale value across the organization.

FDEs Excel Across Three Critical Dimensions

CUSTOMER CENTRICITY
- Deep customer empathy and context
- Co-creation and collaboration
- Trusted advisor and outcome ownership

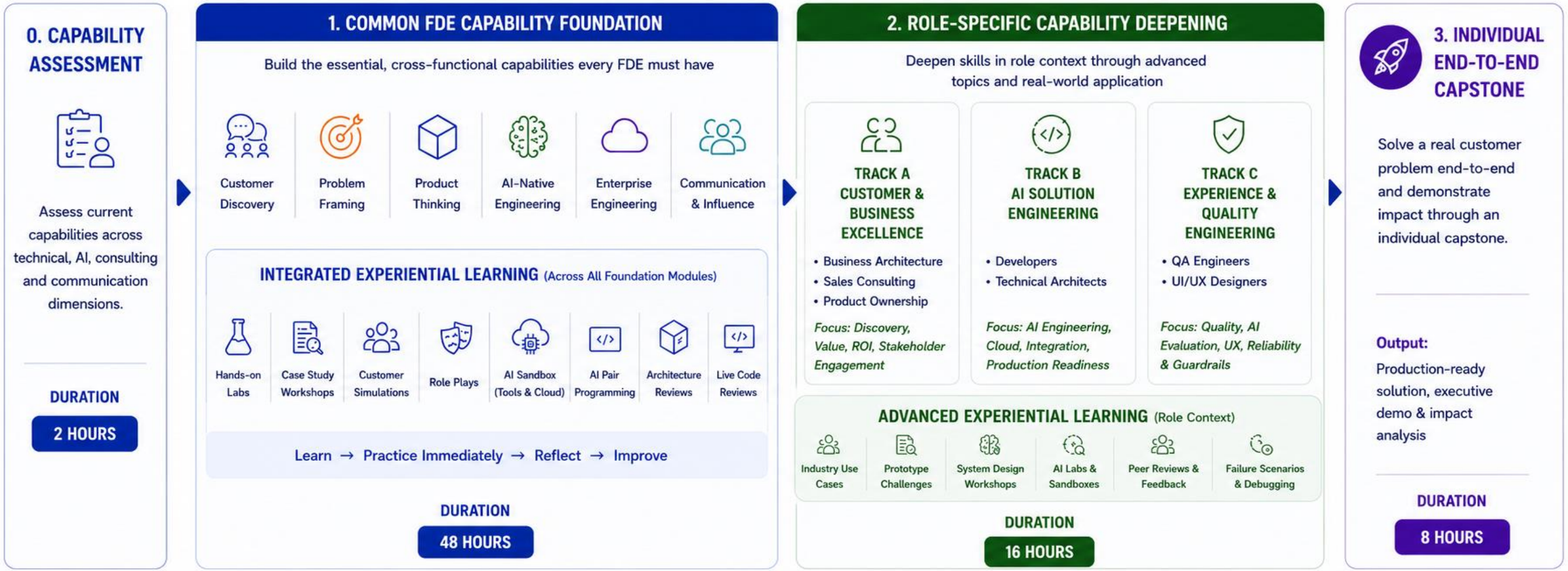
TECHNOLOGY EXCELLENCE
- AI-native engineering and modern architecture
- Enterprise-grade security, scalability and reliability
- Automation, observability and DevSecOps

BUSINESS IMPACT
- Outcome-driven execution
- ROI and KPI focused
- Continuous value realization

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

PROGRAM ARCHITECTURE

Our program is structured across interconnected layers where learning, practice and application are continuously integrated.



A cohesive architecture that develops **AI-native, customer-facing, outcome-driven** Forward Deployed Engineers.

TOTAL PROGRAM DURATION: 74 HOURS

© ***NIIT*** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

ASSUMED COMPETENCIES

The following competencies are assumed for participants to get the most value from the Forward Deployed Engineering program

AI FLUENCY

- Fundamentals of AI/ML
- Large Language Models (LLMs)
- Prompting & Prompt Engineering
- RAG Concepts
- AI Tooling & Ecosystem
- Responsible & Ethical AI

Comfortable using AI tools and understanding how AI solutions are created and applied.

CORE TECHNICAL EXECUTION

- Programming (Java / Python / .NET / Node.js / Go)
- Data Structures & Algorithms
- RESTful APIs & Integration
- SQL / NoSQL Databases
- Basic Frontend Technologies (HTML, CSS, JS)

Able to build, test, and integrate software components independently.

SYSTEM ARCHITECTURE FOUNDATIONS

- Cloud Concepts & Services (AWS / Azure / GCP)
- Distributed Systems Basics
- Microservices Architecture
- Performance & Scalability Fundamentals
- Security Fundamentals

Understands how systems are architected, deployed and secured at an enterprise scale.

STANDARD ENGINEERING WORKFLOWS

- Agile / Scrum Fundamentals
- Source Control (Git)
- CI/CD Fundamentals
- Jira / Issue Tracking
- DevOps Basics & Best Practices
- Testing & Debugging Practices

Familiar with modern engineering practices and collaborative delivery workflows.

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

READINESS ASSESSMENT

A readiness check to baseline participants current capabilities before starting the journey.

WHAT THE ASSESSMENT EVALUATES

1. TECHNICAL READINESS

Ability to analyze technical problems and design practical solutions.

- Problem breakdown & approach
- Architecture & design thinking
- APIs, data & integration understanding
- Cloud & tooling awareness
- Engineering best practices

2. AI READINESS

Ability to identify AI opportunities and apply AI-native concepts.

- AI opportunity identification
- AI approach selection (RAG, Agentic, Fine-tune, etc.)
- Prompt engineering & context awareness
- Responsible AI awareness

3. CONSULTING READINESS

Ability to understand business context and recommend value-driven solutions.

- Discovery & problem framing
- Business understanding
- Solution prioritization
- ROI & success metrics thinking
- Communication of recommendations

4. BEHAVIORAL READINESS

Key behaviors essential for being an effective customer-facing engineer.

- Customer orientation
- Problem solving
- Ownership
- Learning agility
- Curiosity
- Collaboration
- Communication
- Adaptability
- Executive presence
- Decision making under ambiguity

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

COMMON FOUNDATION

Building core capabilities every Forward Deployed Engineer must possess

THE FOUNDATION JOURNEY (48 HOURS)

1. CUSTOMER DISCOVERY & BUSINESS ACUMEN
6 HOURS

- Understand customer context
- Discover & frame the right problems
- Prioritize opportunities
- Communicate & align stakeholders
- Drive business outcomes

2. PRODUCT THINKING & VALUE ENGINEERING
8 HOURS

- Design solutions with customer value
- Prioritize use cases & define MVP
- Build business case
- ROI modelling, KPIs & success metrics
- **Financial metrics & value engineering (NPV, IRR, Payback)**
- Cost–benefit analysis & financial impact assessment
- Token economics & model selection
- Build vs Buy vs Fine-tune decisions
- AI benchmarking & value tracking

3. AI-NATIVE ENGINEERING
14 HOURS

- AI-assisted SDLC
- Claude Code, Cursor, Copilot & Codex
- Prompt & context engineering
- RAG, Vector DBs, MCP, Agentic AI
- AI evaluation, benchmarking & guardrails

4. ENTERPRISE ENGINEERING
14 HOURS

- Cloud platforms (AWS, Azure, GCP)
- APIs, Integration & Kafka
- SQL/NoSQL & Data Modelling
- CI/CD, DevSecOps
- Security, Observability & Production Architecture

5. PRODUCTION READINESS & DELIVERY EXCELLENCE
6 HOURS

- Prototype-driven engagement
- AI-assisted testing & quality
- Deploy, monitor & optimize
- Measure impact & adoption
- Continuous improvement & benchmarking

OUTCOME: Build a common FDE operating model and capability baseline for all engineers before progressing to role-specific capability tracks.

INTEGRATED EXPERIENTIAL LEARNING

Rather than scheduling separate lab sessions, the following learning experiences are integrated into every capability pillar:

- Enterprise AI Sandbox
- Claude Code, Cursor, GitHub Copilot & Codex
- Hands-on Engineering Labs
- Industry Case Studies
- Customer Discovery Simulations
- Role Plays & Consulting Exercises
- AI Pair Programming
- Architecture Reviews
- Code Reviews
- Prototype Challenges
- Peer Feedback & Reflection

CAPABILITY OUTCOMES

At the end of the Common Foundation, every participant will be able to:

- Discover and frame customer problems using a consultative approach.
- Translate business needs into prioritized AI solution opportunities.
- Evaluate business value through ROI, KPIs, **financial metrics,** token economics, and adoption metrics.
- Leverage AI-native engineering tools and modern development practices.
- Design enterprise-ready, secure, scalable, and production-grade solutions.
- Engage confidently with customers through prototypes, technical storytelling, and solution demonstrations.
- Build a shared FDE operating model before progressing to specialized capability tracks.

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

COMMON FOUNDATION

PILLAR	KEY LEARNING OUTCOMES	KEY TOPICS COVERED	LEARNING EXPERIENCES (EMBEDDED ACROSS MODULES)	DURATION (HOURS)
1. CUSTOMER DISCOVERY & BUSINESS ACUMEN	• Understand enterprise business context and customer journeys • Discover, frame and prioritize the right problems • Communicate insights and build stakeholder alignment	• FDE mindset & operating model • Stakeholder discovery & consultative questioning • Pain-point & opportunity mapping • Problem framing & prioritization frameworks • Customer communication & executive storytelling • Business outcome orientation	Customer Simulations Consulting Exercises Role Plays Reflection & Feedback Case Studies	6 HOURS
2. PRODUCT THINKING & VALUE ENGINEERING	• Apply product thinking to design valuable AI solutions • Build business cases and quantify value • Define success through metrics and economics	• Product mindset & solution thinking • Design Thinking & user journey mapping • Use-case prioritization & MVP definition • Business case development & ROI modelling • KPI definition & success metrics • Token economics & AI cost considerations • AI model/tool selection & benchmarking	Hands-on Labs Prototype Challenges Business Case Workshop Peer Discussions Value Modeling Exercises	8 HOURS
3. AI-NATIVE ENGINEERING	• Leverage AI-native tools and techniques across the development lifecycle • Engineer intelligent solutions using modern AI patterns • Ensure responsible and effective use of AI	• AI-assisted SDLC (AI-SDLC) • Generative AI & LLM fundamentals • Claude Code, GitHub Copilot, Cursor & Codex • Prompt Engineering & Context Engineering • RAG fundamentals, Vector Databases & MCP • Agentic AI concepts & orchestration • AI evaluation, benchmarking & guardrails	AI Sandbox & Tooling AI Pair Programming Prompt Labs Model Evaluation Labs Hands-on Engineering Labs	14 HOURS
4. ENTERPRISE ENGINEERING	• Design and build secure, scalable and integrated enterprise solutions • Apply cloud-native and modern engineering practices • Ensure reliability, security and operational excellence	• Cloud-native engineering (AWS, Azure, GCP) • REST, GraphQL & gRPC APIs • SQL & NoSQL, Data Modeling • Enterprise integration patterns • Kafka & event-driven architectures • DevSecOps, CI/CD & release management • Security, Observability & Monitoring • Scalability, reliability & production architecture	Hands-on Labs Integration Labs Architecture Reviews DevOps Simulations Code Reviews	14 HOURS
5. PRODUCTION READINESS & DELIVERY EXCELLENCE	• Deliver production-ready solutions that create real business impact • Demonstrate, measure and optimize customer value • Drive adoption and continuous improvement	• Solution architecture reviews • Prototype-driven customer engagement • AI-assisted testing & quality engineering • Production deployment & release management • Monitoring, optimization & resilience • Customer demos & stakeholder communication • Business impact measurement & adoption metrics • Continuous AI benchmarking & optimization	End-to-End Labs Impact Measurement Workshops Deployment Simulations Retrospectives & Continuous Improvement Customer Demo Practice	6 HOURS

TOTAL DURATION | 48 HOURS

BUSINESS VALUE LENS APPLIED ACROSS ALL PILLARS

ROI Modelling • KPI Definition • Token Economics • AI Cost Optimization • Model Benchmarking • Adoption Metrics • Customer Outcomes

This foundation ensures a common operating model and capability baseline for all FDEs before progressing to role-specific deepening tracks.

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

ROLE-SPECIFIC CAPABILITY DEEPENING

After completing the Common FDE Capability Foundation, learners deepen capabilities based on their primary role as Forward Deployed Engineers (FDEs) to deliver greater customer impact and business value.

DURATION: 16 HOURS

	TRACK 1 CUSTOMER & PRODUCT EXCELLENCE (Business Architects, Sales)	TRACK 2 AI SOLUTION ENGINEERING (Developers, Technical Architects)	TRACK 3 EXPERIENCE & QUALITY ENGINEERING (QA, UI/UX)
FOCUS AREAS	• Advanced customer discovery & stakeholder alignment • Industry & domain deep dive • Value hypothesis & outcome mapping • AI opportunity assessment & prioritization frameworks • Business case development (ROI, NPV, Payback) • AI adoption & change management • Solution positioning & competitive differentiation • Executive storytelling & influence	• Advanced GenAI application development • RAG, Agentic AI & orchestration patterns • Model fine-tuning, evaluation & guardrails • Enterprise data integration & knowledge systems • Scalable architecture & microservices design • Cloud-native development (AWS / Azure / GCP) • Advanced DevSecOps & CI/CD strategies • Observability, performance & cost optimization	• AI UX design & human-centered AI experiences • Conversational design & interaction patterns • AI usability, accessibility & inclusivity • AI quality strategy & test planning • AI test automation & evaluation frameworks • Non-functional testing (performance, security, reliability) • AI safety, bias & robustness validation • Continuous quality & feedback loops
CAPABILITY OUTCOME	FDEs who can lead customer engagements, identify high-value AI opportunities and build compelling business cases that drive adoption and measurable impact.	FDEs who can design, build and operate secure, scalable and AI-native solutions that integrate seamlessly with enterprise systems and deliver production impact.	FDEs who can ensure exceptional user experience, trust, quality and reliability of AI-powered solutions across the entire lifecycle.
INTEGRATED LABS (EXAMPLES)	LAB EXAMPLE 1 Customer Discovery & Opportunity Prioritization Conduct a simulated discovery workshop, identify opportunities and build a prioritized AI opportunity backlog. LAB EXAMPLE 2 Business Case & ROI Simulation Build a full business case with ROI, adoption metrics and executive pitch for a selected AI solution.	LAB EXAMPLE 1 Build an AI-Powered Solution (End-to-End) Build and deploy a RAG/agentic application using enterprise data with observability and guardrails. LAB EXAMPLE 2 Enterprise Integration & Deployment Integrate the solution with enterprise systems (API, data store, event streams) and deploy using CI/CD with monitoring.	LAB EXAMPLE 1 AI UX & Usability Workshop Design and prototype an AI experience, validate user journeys and conduct usability testing. LAB EXAMPLE 2 AI Quality Assurance Lab Design test cases for an AI solution, automate tests and evaluate for safety, bias, reliability and performance.

Outcome: Role-specific depth, stronger applied capability and confidence to deliver measurable customer impact as Forward Deployed Engineers.

END-TO-END CAPSTONE

SAMPLE CAPSTONE PROBLEM STATEMENTS

Participants select one challenge and own the end-to-end solution independently.

1. AI-POWERED CUSTOMER SUPPORT & SERVICE OPERATIONS

Build an AI solution that improves issue resolution, knowledge retrieval, routing and customer experience.

2. ENTERPRISE KNOWLEDGE ASSISTANT

Build an enterprise knowledge assistant with secure retrieval, source citation, permissions and adoption analytics.

3. INTELLIGENT DOCUMENT-TO-DECISION PLATFORM

Build an AI workflow that ingests documents, extracts, validates and routes for approvals with audit trail.

4. IT SERVICE DESK AUTOMATION

Build an AI solution for ticket classification, recommendations, automation, knowledge suggestions and SLA tracking.

CAPSTONE LIFECYCLE

Participants must complete all stages of the FDE lifecycle independently for their chosen challenge.

1 CUSTOMER DISCOVERY
- Stakeholder map
- Problem interviews
- Pain points
- Opportunity assessment

2 PROBLEM FRAMING
- Problem statement
- Root cause analysis
- Success metrics
- Scope & assumptions

3 BUSINESS CASE & ROI
- Value hypothesis
- ROI model
- KPIs & impact
- 90-day value roadmap

4 SOLUTION ARCHITECTURE
- Solution approach
- Architecture diagram
- Tech choices & rationale
- Security & compliance

5 AI-NATIVE ENGINEERING
- Data pipeline / RAG / Agents (as applicable)
- Model / prompt design
- APIs & integrations
- Core functionality build

6 INTEGRATION
- Enterprise system integration
- Workflow / orchestration
- Error handling
- Data & API contracts

7 TESTING & VALIDATION
- Test strategy
- Functional testing
- AI evaluation
- Guardrails & safety
- Quality metrics

8 DEPLOYMENT & OBSERVABILITY
- Cloud deployment
- CI/CD
- Monitoring & logs
- Alerts & dashboards
- Cost optimization

9 CUSTOMER DEMONSTRATION
- Solution demo
- Value realization
- User feedback
- Future enhancements

10 PRODUCTION READINESS
- Runbook
- Security checklist
- Operational checklist
- Risk assessment
- 90-day rollout plan

EVALUATION RUBRIC (TOTAL: 100 POINTS)

EVALUATION DIMENSION	DESCRIPTION	WEIGHTAGE	WHAT IS ASSESSED
1. CUSTOMER & BUSINESS IMPACT	Understanding of customer, problem clarity, business value and ROI.	15%	• Depth of discovery • Problem framing • ROI & KPIs • Value hypothesis
2. SOLUTION DESIGN & ARCHITECTURE	Quality of solution approach, architecture, tech choices and enterprise readiness.	20%	• Architecture • Tech rationale • Security & compliance • Scalability & reliability
3. ENGINEERING & INTEGRATION	Quality of implementation, integrations, workflows and technical execution.	20%	• Code quality • Integrations • Workflows • Data handling • Error handling
4. TESTING, VALIDATION & RELIABILITY	Robustness of testing strategy, AI evaluation, quality & reliability.	15%	• Test coverage • AI evaluation • Guardrails • Reliability metrics • Risk mitigation
5. DEPLOYMENT & OPERATIONS	Production readiness, observability, monitoring and operational excellence.	10%	• Deployment • CI/CD • Monitoring & alerts • Cost optimization
6. COMMUNICATION & DEMONSTRATION	Clarity of demo, storytelling, user experience and ability to communicate value.	10%	• Demo quality • Value communication • UX • Executive readiness
7. INNOVATION & EXCELLENCE	Creativity, innovation, best practices and overall solution differenttation.	10%	• Innovation • Best practices • Differentiation • Future roadmap
	TOTAL	100%	

INDIVIDUAL DELIVERABLES

- Discovery report, problem statement & opportunity assessment
- Business case, ROI model, KPIs & 90-day roadmap
- Solution architecture document & design rationale
- Working solution (code / workflows / integrations)
- Test strategy, test cases, AI evaluation & quality report
- Deployment evidence, dashboards & observability report
- Customer demo video / deck & value narrative
- Production readiness checklist & runbook

ASSESSMENT APPROACH

Artefact Review	50%
Live Demo & Walkthrough	25%
Technical Deep Dive	15%
Q&A & Defense	10%
TOTAL	100%

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

PROGRAM OUTCOMES

STRONGER CUSTOMER ENGAGEMENT

- Deeper understanding of customer context, challenges and outcomes
- Proactive discovery of high-value opportunities
- Stronger trust and long-term customer relationships

HIGHER QUALITY AI SOLUTIONS

- AI-native, secure and scalable solutions designed with best practices
- Better architecture, data integration and model effectiveness
- Responsible AI with built-in guardrails

FASTER TIME TO VALUE

- Accelerated discovery to solution cycle
- Reusable frameworks, assets and AI components
- CI/CD and automation enable rapid, reliable delivery

IMPROVED ENGINEERING PRODUCTIVITY

- AI-native tools and workflows boost developer productivity
- Reduced rework through quality engineering and automated testing
- Observability and monitoring ensure operational excellence

MEASURABLE BUSINESS IMPACT

- Solutions aligned to KPI's and ROI
- Measurable improvement in efficiency, cost savings and revenue impact
- Clear reporting on outcomes and business value

FUTURE-READY FDE CAPABILITY

- A scalable pool of AI-native, customer-facing FDEs
- Continuous learning mindset and adaptability
- Strong foundation for future innovation and emerging technologies

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

WHY NIIT?



Most technical training providers begin with **"Let's start teaching."**



We begin with **"Let's understand who your engineers are likely to become."**



We do not deliver **FDE training programs. We build your capability pipeline.**



WE BUILD ENTERPRISE-READY FORWARD DEPLOYED ENGINEERS.

© **NIIT** *All Information within this document is Intellectual property & cannot be shared or used without the permission of NIIT Ltd.*

THANK YOU

Together, let's build the next generation of **Forward Deployed Engineers.**

Let's connect
and create impact together.

